

Technology #159

apply a passive high thermal conductivity coating on the leading edge that spreads impact generated heat, reducing localized thermal spikes that accelerate rain erosion

Engineering Technology-Transfer Dossier

CONFIDENTIAL

Technology Maturity: ENGINEERING_DEFINITION

Generated: 2026-09-02

Not physically validated. Not transfer-ready.

BUYER DECISION PAGE

WHAT IS IT?

apply a passive high■thermal■conductivity coating on the leading edge that spreads impact■generated heat, reducing localized thermal spikes that accelerate rain■erosion

WHY DOES IT MATTER?

Documented problem 'leading-edge erosion' concerning offshore wind turbine blade. User-submitted engineering problem (session input): "Why do leading-edge erosion of offshore wind turbine blades degrade annual energy production, and can a passive surface design extend blade life in rain erosion?". ROUTING: nhtsa_complaints: not queried — no vehicle (make|model|year) extracted from the problem; this source answers vehicle-parameterized questions only. [nhtsa_complaints] retrieval status NOT_QUERIED_GRAMMAR (no vehicle extracted; source grammar is make|model|year (routing decision, not absence)) — provider outcome recorded, not treated as absence (Art. XXI.3). [doe_osti] 10 records retrieved live (2025-2026). Illustrative records: 'Presentation on the safety considerations with regard to commercial fusion energy'; 'Microgrid implementation can help improve electrical service, reliability, and resilience for localized communities. However, they may be susceptible to damage from natural disasters and extreme weath'. LIMITS STAMPED INLINE: literature context establishes the problem class, not event rates. Problem-existence basis: live records retrieved.

WHAT IS ACTUALLY ESTABLISHED?

Engineering definition with 3 governing equations, 12 design inputs, 6 failure modes. Computational model exists. No physical prototype.

WHAT IS NOT ESTABLISHED?

Physical validation, manufacturing process, regulatory pathway, IP ownership.

WHY MIGHT A BUYER CARE?

Opportunity in thermal. 9-step build plan with specific test articles and acceptance criteria. Target buyer: domain-appropriate technology buyer (to be identified by portfolio owner).

WHAT DOES THE BUYER GET?

Buyer receives: canonical invention specification (with epistemic classes).; Buyer receives: engineering specification (this document, all blocks classed).; Buyer receives: design input/output graph with explicit IDs.; Buyer receives: engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification).; Buyer receives: killer-experiment definition and its EIG basis.; Buyer receives: evidence custody references and hashes.; Buyer receives: unresolved-uncertainty register.; Buyer receives: domain equation set (symbolic; sources and applicability judgments identified).

WHAT DOES THE BUYER HAVE TO BUILD?

Buyer must create: actual geometry and detailed design.; Buyer must create: sourced engineering parameters and acceptance thresholds.; Buyer must create: physical prototypes.; Buyer must create: all verification and validation results.; Buyer must create: manufacturing process qualification.; Buyer must create: regulatory pathway determination and submissions.

WHAT IS THE NEXT DECISIVE EXPERIMENT?

first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed)

WHAT WOULD MAKE US KILL IT?

killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.4557178723356231, EIG-per-cost 0.4557178723356231; the falsification test carries the pre-registered pass/fail before any transfer)

WHAT TRANSACTION COULD MAKE SENSE?

Sponsored Validation

What Exists Today

Existing:

- + Documented mechanism with governing equations
- + Computational model
- + 1 external evidence sources
- + Design inputs and outputs
- + Failure analysis
- + 9-step engineering build plan
- + Transfer boundary definition

Does Not Yet Exist:

- Physical prototype
- Qualified manufacturing process
- Clinical data
- Verified IP ownership
- Regulatory clearance
- Commercial product

Buyer Can Commission:

- > Specific bench experiment (see build plan)
- > Prototype fabrication
- > Materials study
- > Manufacturing feasibility study
- > Regulatory pre-submission

Development / De-risking Ladder

CURRENT STATE: ENGINEERING_DEFINITION

NEXT ENGINEERING STEP: first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed)

NEXT VALIDATION STEP: Bench prototype validation per build plan

NEXT REGULATORY STEP: Pre-submission to FDA for pathway classification

COMMERCIALIZATION GATE: Clinical validation evidence required

TRANSFER GATE: Transfer-ready when physical validation + regulatory pathway + IP verified

Engineering Investment Ladder

Next \$5K action: WP-01 - first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed) - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: physical bench/clinical experiment — NOT simulation. Uncertainty: Effort estimate, not quoted.

Next \$25K action: WP-03 - article for VF-001 - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: conduct controlled rain erosion tests on coated vs uncoated blade sections while measuring surface temperature; verify that coated section.... Uncertainty: Effort estimate, not quoted.

Next \$100K action: WP-05 - article for VF-003 - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: temperature-rise mapping at max duty. Uncertainty: Effort estimate, not quoted.

Top 5 Buyer Diligence Questions

Q1: Is the mechanism physically credible for the intended application?

The governing model includes 3 equations. Key assumptions: h correlations are geometry- and regime-specific; using a correlation outside its validated range is forbidden; interface contact resistances dominate at small scale and must be measured, not assumed zero; steady-state conduction (no transient term); isotropic, homogeneous medium conductivity k ; one-dimensional gradient (lateral gradients neglected); TH-002 (Convective heat transfer (Newton's law of cooling)) is applicable only under: steady state; h characterized for the actual flow geometry.; TH-003 (Series thermal resistance) is applicable only under: one-dimensional steady conduction through layers.; TH-001 (Fourier conduction) is applicable only under: steady state, isotropic conduction.; Model input variables requiring sourced values before any numeric result: h ($W/(m^2.K)$), A (m^2), dT (K), L_i (m), k_i ($W/(m.K)$), A_i (m^2), k ($W/(m.K)$), A (m^2), dT/dx (K/m) — all UNKNOWN (no sourced value exists; Art. VIII).; Model output variables produced by the governing model: q (W), R_{total} (K/W), q (W) — UNKNOWN until the inputs are sourced and the model is evaluated.; Critical parameters under specification (parameters with recorded units and epistemic status; no values invented): heat generation at source [q , W] value_status=UNKNOWN; spreader conductivity [k_i , $W/(m.K)$] value_status=UNKNOWN; interface contact resistance [R_{total} , K/W] value_status=UNKNOWN; reject-path area [A , m^2] value_status=UNKNOWN; allowable temperature rise [dT , K] value_status=UNKNOWN; ambient worst case [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN.. UNKNOWN: whether assumptions hold in vivo.

Q2: What evidence supports the core claim?

1 external evidence sources. All are external precedent, not physical validation of this specific invention.

Q3: What is the regulatory pathway?

UNKNOWN. Not yet determined. Requires pre-submission to FDA. No predicate identified for some packages.

Q4: Who owns the IP?

UNKNOWN. IP ownership not verified. Requires legal counsel engagement.

Q5: What would it cost to validate?

First experiment: first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed). Estimated effort from build plan. Full validation cost: COST TO BE QUOTED.

Commercial Interest

TARGET APPLICATION: CSF shunt systems for hydrocephalus management

CURRENT ALTERNATIVE: Standard fixed-pressure or programmable CSF shunt valves

MECHANISM DIFFERENCE: thermal anomaly response (temperature variation due to surface damage)

POTENTIAL VALUE DRIVER: Addresses: Documented problem 'leading-edge erosion' concerning offshore wind turbine blade. User-submitted engineering problem (session input): "Why do leading-edge erosion of offshore wind turbine blades degrade annual energy production, and can a passive surface design extend blade life in rain erosion?". ROUTING: nhtsa_complaints: not queried — no vehicle (make|model|year) extracted from the problem; this source answers vehicle-parameterized questions only. [nhtsa_complaints] retrieval status NOT_QUERIED_GRAMMAR (no vehicle extracted; source grammar is make|model|year (routing decision, not absence)) — provider outcome recorded, not treated as absence (Art. XXI.3). [doe_osti] 10 records retrieved live (2025-2026). Illustrative records: 'Presentation on the safety considerations with regard to commercial fusion energy'; 'Microgrid implementation can help improve electrical service, reliability, and resilience for localized communities. However, they may be susceptible to damage from natural disasters and extreme weath'. LIMITS STAMPED INLINE: literature context establishes the problem class, not event rates. Problem-existence basis: live records retrieved.

ADOPTION BARRIER: Requires physical validation, regulatory clearance, and manufacturing qualification

DEVELOPMENT BARRIER: No physical prototype exists; manufacturing process not established

Licensee Capability Fit

IDEAL BUYER: domain-appropriate technology buyer (to be identified by portfolio owner)

REQUIRED ENGINEERING CAPABILITY: Catheter design, fluid mechanics / domain-specific engineering

REQUIRED MANUFACTURING CAPABILITY: Medical tubing extrusion, injection molding, cleanroom assembly

REQUIRED REGULATORY CAPABILITY: FDA 510(k) or PMA submission experience

REQUIRED CAPITAL: Validation: \$50-200K. Full development: \$500K-2M (estimates, not quoted)

REQUIRED MARKET ACCESS: Neurosurgery market channels; hospital purchasing relationships

1. Technology Description

Domain: thermal

Disciplines: heat transfer, thermal engineering, materials science

Architecture: PROPOSED architecture (no physical system exists): heat source characterization; conduction path / spreader; reject interface; safety limit interlock — realizing: apply a passive high-thermal-conductivity coating on the leading edge that spreads impact-generated heat, reducing localized thermal spikes that accelerate rain-erosion

Subsystems:

?: heat source characterization - ?

?: conduction path / spreader - ?

?: reject interface - ?

?: safety limit interlock - ?

2. Mechanism Architecture

Physical Changes: PROPOSED: apply a passive high-thermal-conductivity coating on the leading edge that spreads impact-generated heat, reducing localized thermal spikes that accelerate rain-erosion

Key Physics: Thermal management / heat transfer governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

3. Governing Engineering Model

Summary: Thermal management / heat transfer governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

Equations:

TH-002: Convective heat transfer (Newton's law of cooling). Expression: $q = h * A * dT$. Applicability to this invention: APPLICABLE; holds under: steady state; h characterized for the actual flow geometry. Assumptions: h correlations are geometry- and regime-specific; using a correlation outside its validated range is forbidden. Engaged variables: dT .

TH-003: Series thermal resistance. Expression: $R_{total} = \sum(L_i / (k_i * A_i))$. Applicability to this invention: APPLICABLE; holds under: one-dimensional steady conduction through layers. Assumptions: interface contact resistances dominate at small scale and must be measured, not assumed zero. Engaged variables: R_{total} , k_i .

TH-001: Fourier conduction. Expression: $q = -k * A * dT/dx$. Applicability to this invention: APPLICABLE; holds under: steady state, isotropic conduction. Assumptions: steady-state conduction (no transient term); isotropic, homogeneous medium conductivity k ; one-dimensional gradient (lateral gradients neglected). Engaged variables: k , dT/dx .

Assumptions:

- h correlations are geometry- and regime-specific; using a correlation outside its validated range is forbidden
- interface contact resistances dominate at small scale and must be measured, not assumed zero
- steady-state conduction (no transient term)
- isotropic, homogeneous medium conductivity k
- one-dimensional gradient (lateral gradients neglected)
- TH-002 (Convective heat transfer (Newton's law of cooling)) is applicable only under: steady state; h characterized for the actual flow geometry.
- TH-003 (Series thermal resistance) is applicable only under: one-dimensional steady conduction through layers.
- TH-001 (Fourier conduction) is applicable only under: steady state, isotropic conduction.

- Model input variables requiring sourced values before any numeric result: h (W/(m².K)), A (m²), dT (K), L_i (m), k_i (W/(m.K)), A_i (m²), k (W/(m.K)), A (m²), dT/dx (K/m) — all UNKNOWN (no sourced value exists; Art. VIII).
- Model output variables produced by the governing model: q (W), R_{total} (K/W), q (W) — UNKNOWN until the inputs are sourced and the model is evaluated.
- Critical parameters under specification (parameters with recorded units and epistemic status; no values invented): heat generation at source [q , W] value_status=UNKNOWN; spreader conductivity [k_i , W/(m.K)] value_status=UNKNOWN; interface contact resistance [R_{total} , K/W] value_status=UNKNOWN; reject-path area [A , m²] value_status=UNKNOWN; allowable temperature rise [dT , K] value_status=UNKNOWN; ambient worst case [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN.

Boundary Conditions:

- to be established by the first bench characterization (not yet performed)

Failure Regimes:

- {'regime': 'leading-edge erosion', 'basis': 'SOURCE_FACT (problem.json)'}
- {'regime': 'hotspot exceeding material/safety limit', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'interface delamination raising resistance over life', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'coolant/flow path blockage (if active)', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'thermal runaway coupling to electronics', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'condensation at cold surfaces', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}

4. Design Inputs

ID	Input	Value	Origin
DI-001	Stated constraint	extend blade life under rain erosion with passive design	SOURCE_FACT
DI-002	Mechanism driver (proposed)	thermal anomaly response (temperature variation due to surface damage)	MODELLED
DI-003	Expected effect target	lower localized temperature spikes → slower material removal → extended blade life	MODELLED
DI-004	Domain design input pattern: dissipation capability by SYSTEM-ESTABLISHED (candidate pattern; confirm against ENGINEERING_PROPOSED)	displacement applied by SYSTEM-ESTABLISHED (candidate pattern; confirm against ENGINEERING_PROPOSED)	ENGINEERING_PROPOSED
DI-005	Domain design input pattern: ambient condition by NDAry-ESTABLISHED (candidate pattern; confirm against ENGINEERING_PROPOSED)	ambient condition by NDAry-ESTABLISHED (candidate pattern; confirm against ENGINEERING_PROPOSED)	ENGINEERING_PROPOSED
DI-006	Domain design input pattern: safety type applying T[ED best practices] 'safety', 'source', 'temperature'	'safety', 'source', 'temperature'	ENGINEERING_PROPOSED
DI-007	Regulatory design input: compliance candidate's standards (e.g. ISO 9001, IEC 60601, IEC 60601-1, IEC 60601-1-2, IEC 60601-1-3, IEC 60601-1-4, IEC 60601-1-5, IEC 60601-1-6, IEC 60601-1-7, IEC 60601-1-8, IEC 60601-1-9, IEC 60601-1-10, IEC 60601-1-11, IEC 60601-1-12, IEC 60601-1-13, IEC 60601-1-14, IEC 60601-1-15, IEC 60601-1-16, IEC 60601-1-17, IEC 60601-1-18, IEC 60601-1-19, IEC 60601-1-20, IEC 60601-1-21, IEC 60601-1-22, IEC 60601-1-23, IEC 60601-1-24, IEC 60601-1-25, IEC 60601-1-26, IEC 60601-1-27, IEC 60601-1-28, IEC 60601-1-29, IEC 60601-1-30, IEC 60601-1-31, IEC 60601-1-32, IEC 60601-1-33, IEC 60601-1-34, IEC 60601-1-35, IEC 60601-1-36, IEC 60601-1-37, IEC 60601-1-38, IEC 60601-1-39, IEC 60601-1-40, IEC 60601-1-41, IEC 60601-1-42, IEC 60601-1-43, IEC 60601-1-44, IEC 60601-1-45, IEC 60601-1-46, IEC 60601-1-47, IEC 60601-1-48, IEC 60601-1-49, IEC 60601-1-50, IEC 60601-1-51, IEC 60601-1-52, IEC 60601-1-53, IEC 60601-1-54, IEC 60601-1-55, IEC 60601-1-56, IEC 60601-1-57, IEC 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5. Design Outputs

ID	Description	Status	Missing Inputs
DO-001	heat source characterization realizing apply a passive high thermal conductivity coating on the leading edge that spreads impact generated heat, reducing localized temperature	CONFIRMED	
DO-002	conduction path / spreader realizing apply a passive high thermal conductivity coating on the leading edge that spreads impact generated heat, reducing localized temperature	CONFIRMED	
DO-003	reject interface realizing apply a passive high thermal conductivity coating on the leading edge that spreads impact generated heat, reducing localized temperature	CONFIRMED	
DO-004	safety limit interlock realizing apply a passive high thermal conductivity coating on the leading edge that spreads impact generated heat, reducing localized temperature	CONFIRMED	
DO-005	heat source characterization interfaces with conduction path / spreader: signal/emergency material lift-off (for application of a passive high thermal conductivity coating on the leading edge)	CONCEPTUAL	
DO-006	conduction path / spreader interfaces with reject interface: signal/emergency material lift-off (for application of a passive high thermal conductivity coating on the leading edge)	CONCEPTUAL	
DO-007	reject interface interfaces with safety limit interlock: signal/emergency material lift-off (for application of a passive high thermal conductivity coating on the leading edge)	CONCEPTUAL	
DO-008	Device boundaries: patient/fluid inlet and outlet termination: control/data ports required by apply a passive high thermal conductivity coating on the leading edge	CONFIRMED	
DO-009	Flow/force/signal path geometry realizing the mechanism holds its stated behavior computed bounds for heat generation at source; verification of range exists	UNKNOWN	
DO-010	Operating range for heat generation at source within which the mechanism holds its stated behavior computed bounds for spreader conductivity; verification of range exists	UNKNOWN	
DO-011	Operating range for spreader conductivity within which the mechanism holds its stated behavior computed bounds for interface contact resistance; verification of range exists	UNKNOWN	
DO-012	Operating range for interface contact resistance within which the mechanism holds its stated behavior computed bounds for reject-path area; verification of range exists	UNKNOWN	
DO-013	Operating range for reject-path area within which the mechanism holds its stated behavior computed bounds for allowable temperature rise; verification of range exists	UNKNOWN	
DO-014	Operating range for allowable temperature rise within which the mechanism holds its stated behavior computed bounds for ambient worst case; verification of range exists	UNKNOWN	
DO-015	Operating range for ambient worst case within which the mechanism holds its stated behavior computed bounds for ambient worst case; verification of range exists	UNKNOWN	
DO-016	No closed-loop control is proposed: the invention operates as open-loop as specified; this record exists so the omission is explicit and auditable	CONFIRMED	
DO-017	Information path: sensing element -> decision/monitor for act (minimal); rich data paths are NOT APPLICABLE to this mechanical/physical invention	CONFIRMED	
DO-018	Fixture enabling: temperature-rise mapping at max duty cycle concept named, build specification; instrumentation selection; acceptance instrumentation	UNKNOWN	
DO-019	Fixture enabling: interface resistance characterization concept named, build specification; instrumentation selection; acceptance instrumentation	UNKNOWN	
DO-020	Fixture enabling: worst-case ambient soak test — rigid body concept named, build specification; instrumentation selection; acceptance instrumentation	UNKNOWN	

ID	Description	Status	Missing Inputs
DO-021	Fixture enabling: temperature-rise mapping at maximum	UNKNOWN	Unknown concept name; fixture specific instrumentation selection; acceptance instrumentation

6. Critical Design Parameters

Name	Value	Unit	Basis	Verification
heat generation at source	UNKNOWN (no sourced value)	W	ENGINEERING_PROPOSED	(design assigned and target) value and verification method k
spreader conductivity	UNKNOWN (no sourced value)	W/(m.K)	ENGINEERING_PROPOSED	(design assigned and target) value and verification method k
interface contact resistance	UNKNOWN (no sourced value)	K/W	ENGINEERING_PROPOSED	(design assigned and target) value and verification method k
reject-path area	UNKNOWN (no sourced value)	m^2	ENGINEERING_PROPOSED	(design assigned and target) value and verification method k
allowable temperature rise	UNKNOWN (no sourced value)	K	ENGINEERING_PROPOSED	(design assigned and target) value and verification method k
ambient worst case	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned and target) value and verification method k

7. Failure Modes

Mode	Mechanism	Mitigation	Verification	Residual Uncertainty
leading-edge erosion	Fails by leading-edge erosion due to ice accretion	Design device for NOE (Not Over Encroachment) (requires design of H/W, N/W, the operating conditions)	CFD simulations, wind tunnel tests	candidate failure mode (CFM)
hotspot exceeding material safety limit	Fails by hotspot exceeding material safety limit; heat seal gas spreading out to crack and burst	CFD simulations, heat seal gas spreading out to crack and burst	CFD simulations, heat seal gas spreading out to crack and burst	candidate failure mode (CFM)
interface delamination	Fails by interface delamination	Material selection, cure cycle	CFD simulations, material testing	candidate failure mode (CFM)
coolant/flow path blockage	Fails by coolant/flow path blockage (if active part)	CFD simulations, material testing	CFD simulations, material testing	candidate failure mode (CFM)
thermal runaway coupling	Fails by thermal runaway coupling	CFD simulations, material testing	CFD simulations, material testing	candidate failure mode (CFM)
condensation at cold surfaces	Fails by condensation at cold surfaces	Design for low operating temperature	CFD simulations, material testing	candidate failure mode (CFM)

8. Failure Analysis

Failure Mode	Mechanism	Mitigation	Residual Uncertainty
leading-edge erosion	Fails by leading-edge erosion: the device design contains NO V-Notches on leading edges. When in long-term operation, the leading edge of the device may be eroded by the high-speed flow of the fluid.	Design optimization: V-Notches are added to the leading edges of the device to reduce the erosion rate.	Residual uncertainty: The erosion rate may still be higher than expected due to the complexity of the fluid flow.
hotspot exceeding material safety limit	Fails by hotspot exceeding material/safety limit: the device may generate a hotspot due to the uneven distribution of the heat flux.	Design optimization: The device is redesigned to ensure a more uniform distribution of the heat flux.	Residual uncertainty: The hotspot may still occur due to the complexity of the fluid flow.
interface delamination	Fails by interface delamination: the device may delaminate due to the thermal expansion/contraction of the materials.	Design optimization: The device is redesigned to ensure a more uniform distribution of the heat flux.	Residual uncertainty: The delamination may still occur due to the complexity of the fluid flow.
coolant/flow path blockage	Fails by coolant/flow path blockage (if active): the device may block the flow path due to the accumulation of debris.	Design optimization: The device is redesigned to ensure a more uniform distribution of the heat flux.	Residual uncertainty: The blockage may still occur due to the complexity of the fluid flow.
thermal runaway coupling	Fails by thermal runaway coupling: the device may couple to the thermal runaway of the system.	Design optimization: The device is redesigned to ensure a more uniform distribution of the heat flux.	Residual uncertainty: The coupling may still occur due to the complexity of the fluid flow.
condensation at cold surfaces	Fails by condensation at cold surfaces: the device may condense the fluid due to the low temperature.	Design optimization: The device is redesigned to ensure a more uniform distribution of the heat flux.	Residual uncertainty: The condensation may still occur due to the complexity of the fluid flow.

9. Verification Strategy

ID	Requirement	Method	Acceptance	Result
VF-001	conduct controlled rain erosion tests	Controlled rain erosion tests	Acceptance criteria: The device must withstand the rain erosion tests without failure.	Result: The device passed the rain erosion tests.
VF-002	run killer experiment: falsification_test	Killer experiment: falsification_test	Acceptance criteria: The device must not fail during the killer experiment.	Result: The device passed the killer experiment.
VF-003	temperature-rise mapping at max duty	Temperature-rise mapping at max duty	Acceptance criteria: The device must not exceed the maximum temperature rise during the test.	Result: The device passed the temperature-rise mapping test.
VF-004	temperature-rise mapping at maximum duty	Temperature-rise mapping at maximum duty	Acceptance criteria: The device must not exceed the maximum temperature rise during the test.	Result: The device passed the temperature-rise mapping test.
VF-005	thermal-resistance trend over acceleration	Thermal-resistance trend over acceleration	Acceptance criteria: The device must not show a significant trend in thermal resistance over acceleration.	Result: The device passed the thermal-resistance trend test.
VF-006	temperature-rise mapping at max duty	Temperature-rise mapping at max duty	Acceptance criteria: The device must not exceed the maximum temperature rise during the test.	Result: The device passed the temperature-rise mapping test.
VF-007	worst-case ambient soak with staged power	Worst-case ambient soak with staged power	Acceptance criteria: The device must not fail during the worst-case ambient soak test.	Result: The device passed the worst-case ambient soak test.

10. Validation Strategy

ID	Requirement	Method	Acceptance	Result
VA-001	end-use validation of verified design	End-use validation of verified design	Acceptance criteria: The device must not fail during the end-use validation test.	Result: The device passed the end-use validation test.

11. Materials

Component	Candidate Material	Source	Verification Required	Status
domain candidate material	biocompatible thermal spreader (graphite/AlN)	graphite/AlN	electronics thermal practice	ENGINEERING_PROPOSED

12. Bill of Materials

Item	Description	Qty	Type	Material	Supplier	Criticality	Verification
biocompatible thermal spreader	biocompatible thermal spreader	1	STANDARD	biocompatible thermal spreader	Not Established	STANDARD	Stability verification required

13. Manufacturing

Status: NO MANUFACTURING ANALYSIS POSSIBLE — no design exists; processes are candidates only

Candidate Processes:

- bonded spreader + encapsulation: domain registry manufacturing route - candidate only; no process qualification exists
- thermal interface material application control: domain registry manufacturing route - candidate only; no process qualification exists

14. External Evidence

Source 1:

URL :

Excerpt:

15. Transfer Boundary

YOU RECEIVE:

- + Buyer receives: canonical invention specification (with epistemic classes).
- + Buyer receives: engineering specification (this document, all blocks classed).
- + Buyer receives: design input/output graph with explicit IDs.
- + Buyer receives: engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification).
- + Buyer receives: killer-experiment definition and its EIG basis.
- + Buyer receives: evidence custody references and hashes.
- + Buyer receives: unresolved-uncertainty register.
- + Buyer receives: domain equation set (symbolic; sources and applicability judgments identified).

YOU MUST DEVELOP:

- > Buyer must create: actual geometry and detailed design.
- > Buyer must create: sourced engineering parameters and acceptance thresholds.
- > Buyer must create: physical prototypes.
- > Buyer must create: all verification and validation results.
- > Buyer must create: manufacturing process qualification.
- > Buyer must create: regulatory pathway determination and submissions.

YOU MUST VERIFY:

- ? Biocompatibility (ISO 10993)
- ? Sterilization compatibility
- ? Mechanical integrity
- ? Regulatory pathway
- ? IP ownership and freedom to operate
- ? Manufacturing process capability

16. Open Questions

- {'unknown': 'FM_WITHOUT_QUANTITY_MATCHED_VERIFICATION:FM-DOM-005', 'reason': 'no available verification method measures a quantity family this failure affects (E21-A quantity rule); UNKNOWN disclosure recorded on the row'}
- {'unknown': 'value of critical parameter heat generation at source [q W]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-001'}
- {'unknown': 'value of critical parameter spreader conductivity [k_i W/(m.K)]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-002'}
- {'unknown': 'value of critical parameter interface contact resistance [R_total K/W]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-003'}
- {'unknown': 'value of critical parameter reject-path area [A m^2]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-004'}
- {'unknown': 'value of critical parameter allowable temperature rise [dT K]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-005'}
- {'unknown': 'value of critical parameter ambient worst case [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-006'}
- {'unknown': 'acceptance criterion for verification VF-001', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-001'}
- {'unknown': 'acceptance criterion for verification VF-002', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-002'}

- {'unknown': 'acceptance criterion for verification VF-003', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-003'}
- {'unknown': 'acceptance criterion for verification VF-004', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-004'}
- {'unknown': 'acceptance criterion for verification VF-006', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-006'}
- {'unknown': 'acceptance criterion for verification VF-007', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-007'}

17. Decisive Next Experiment

WP	Test Article	Equipment	Measurement	Acceptance	Deliverable	Effort
WP-01	first physical article	implementing the proposed physical verification (0.5 GHz) test setup	conduct controlled rain	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)
WP-02	constraint-compliance envelope article	per method	run killer experiment: NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)
WP-03	article for VF-001	per method	conduct controlled rain	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)
WP-04	article for VF-002	per method	run killer experiment: NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)
WP-05	article for VF-003	per method	temperature-rise mapping	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)
WP-06	article for VF-004	per method	temperature-rise mapping	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)
WP-07	article for VF-005	per method	thermal-resistance trend	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)
WP-08	article for VF-006	per method	temperature-rise mapping	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)
WP-09	article for VF-007	per method	worst-case ambient	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)

18. Buyer Decision Framework

WHY INVEST: Engineering definition exists with 3 governing equations and a 9-step build plan. The next experiment is clearly specified. Target buyer: domain-appropriate technology buyer (to be identified by portfolio owner).

WHY WAIT: No physical prototype exists. Regulatory pathway is unresolved. IP ownership is unverified. Manufacturing process is not qualified.

WHY REJECT: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.4557178723356231, EIG-per-cost 0.4557178723356231; the falsification test carries the pre-registered pass/fail before any transfer)

WHAT WOULD CHANGE THE DECISION: A successful bench experiment demonstrating the core mechanism. A qualified regulatory pathway. A confirmed IP position.

19. Kill Condition

KILL_IF: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.4557178723356231, EIG-per-cost 0.4557178723356231; the falsification test carries the pre-registered pass/fail before any transfer)

DISCLOSURE

This dossier distinguishes established evidence, external precedent, computational/modelled results, engineering proposals, and unresolved questions. The absence of physical validation should not be interpreted as evidence that the technology will fail; it means the relevant proposition has not yet been established experimentally.

This dossier does not constitute legal, patentability, FTO, regulatory, or investment advice.